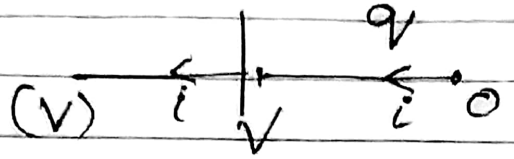
Power & Energy

→ "we will study Power in detail in A.C."



$$\frac{W_{ext}}{q} = V - 0$$

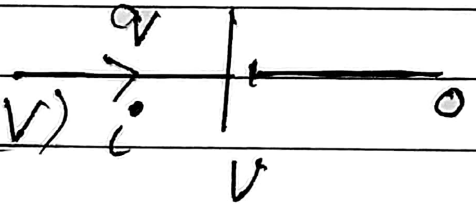
$$W_{ext} = qV$$

Battery is discharging

Energy given by battery = $\Delta U = qV = $

Power given by battery = $\frac{\text{Energy}}{\text{time}} = \frac{qV}{t}$

Power supplied by battery $= iV = Vi$



$$\frac{W_{ext}}{q} = 0 - V$$

Battery is charging

$$W_{ext} = -qV$$

$$\Delta U = -qV$$

$$P = \frac{\Delta U}{t} = \frac{-qV}{t}$$

Power absorbed by battery $= P = -Vi$

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$P = Vi$

$P = i^2 R$

$P = \frac{V^2}{R}$

Power absorbed in Resistor

$P = \frac{\text{Energy}}{\text{time}}$

$P = \frac{dE}{dt}$

$\int dE = \int P\,dt$

$\int dE = \int i^2 R\,dt$

$E = R\int i^2 dt$

if i = constant

Energy absorbed by Resistor

$$\boxed{E = i^2 R t}$$

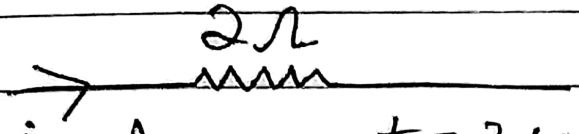
or

$$\boxed{E = \frac{V^2}{R} \times t}$$

or

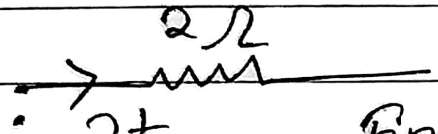
$$\boxed{E = Vi\,t}$$

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$t = 2\,sec$ find Energy

$$E = i^2 R t = 9 \times 2 \times 2 = 36\,J$$



Find E in 2 sec

$$E = \int i^2 R\,dt = \int_0^2 4t^2 \times 2 \times dt$$

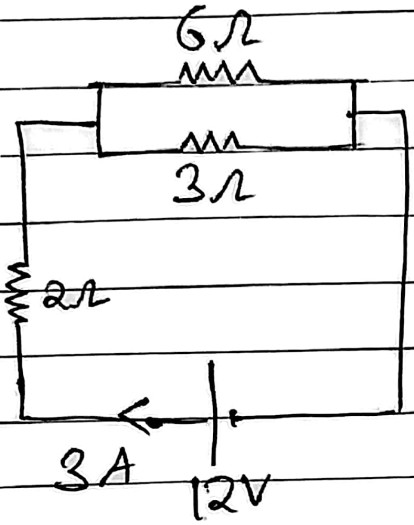
$$= 8\left[\frac{t^3}{3}\right]_0^2 = \frac{8 \times 8}{3} = \frac{64}{3}\,J$$

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Q) In the given Circuit Find

i) Energy spent in 3Ω resister in 3sec

ii) Energy spent in complete circuit in 3sec

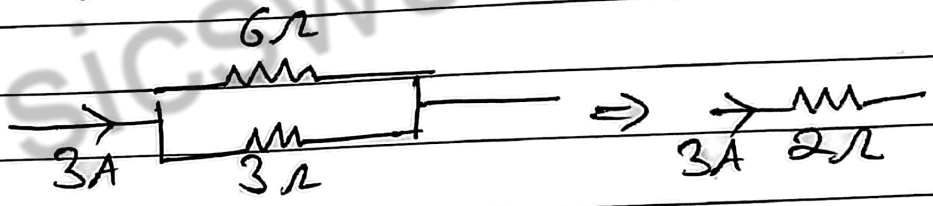


$R_e = 4\Omega$

$V = 12\,V$

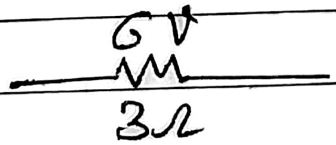
$i = \frac{12}{4} = 3A$

for 3Ω resister



$V = iR = 6V$

So 6V is P.d in both 6Ω & 3Ω



$E = \frac{V^2}{R} \times t = \frac{6 \times 6}{3} \times 3$

$= 36\,J$

For Whole Circuit $V = 12V$ $i = 3A$ $R_e = 4\Omega$

$E = Vit = 12 \times 3 \times 3 = 108\,J$

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Power Rating of an Appliance:



⇒ Resistance of bulb ⇒ $P = \frac{V^2}{R}$

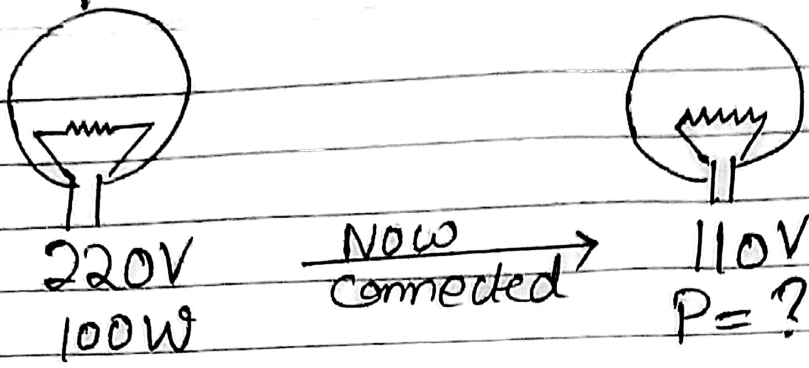
$$R = \frac{V^2}{P} = \frac{(220)^2}{100} = 484\,\Omega$$

⇒ Max current it can take ⇒ $P = Vi$

$$i = \frac{P}{V} = \frac{100}{220} = \frac{5}{11}\,A$$

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100W se chamkega Jab 220V milega



Note: Resistance will be same in both case.

$$R = \frac{V^2}{P}$$

$$= \frac{(220)^2}{100}$$

$$= 484\,\Omega$$

$$P = \frac{V^2}{R} = \frac{(110)^2}{484}$$

$$= \frac{110 \times 110}{484}$$

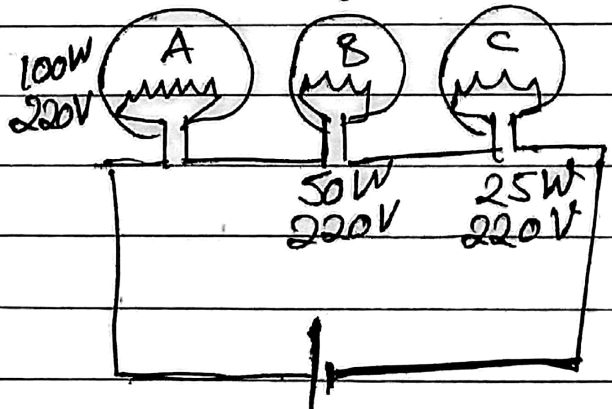
$$= \frac{10}{44} \times 110 = \frac{10 \times 10}{4}$$

$$= 25 \text{ Watt}$$

So it glows dimmer

(Brightness $\propto$ Power)

Q) Which bulb will glow brightest



a) A
b) B
c) C
d) All glow equally bright

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Remember: Indian household supply is 220V | U.S. household supply is 110V

Solution :

Series Combination ⟹ i is same in all

$$R_1 = \frac{(220)^2}{100} \qquad R_2 = \frac{(220)^2}{50} \qquad R_3 = \frac{(220)^2}{25}$$

$$P_1 = i^2 R_1 \qquad P_2 = i^2 R_2 \qquad P_3 = i^2 R_3$$

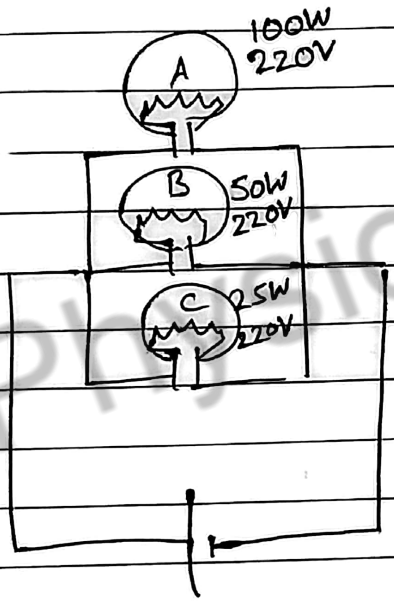
i is same & $R_3 > R_2 > R_1$

$\Rightarrow$ $P_3 > P_2 > P_1$

$\downarrow$

order of brightness

Q)



Which bulb will glow brightest

a) A
b) B
c) C
d) All glow equally bright

Solution:

Parallel combination :→ V is same in all

$$R_1 = \frac{(220)^2}{100} \qquad R_2 = \frac{(220)^2}{50} \qquad R_3 = \frac{(220)^2}{25}$$

$$P_1 = \frac{V^2}{R_1} \qquad P_2 = \frac{V^2}{R_2} \qquad P_3 = \frac{V^2}{R_3}$$

V is same & $R_3 > R_2 > R_1$

$\Rightarrow$ $P_1 > P_2 > P_3$

order of brightness ↑

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IIT 2010

Q) Resistance of 100W, 60W & 40W bulbs are R_{100}, R_{60} & R_{40}

a) $\frac{1}{R_{100}} = \frac{1}{R_{60}} + \frac{1}{R_{40}}$ b) $R_{100} = R_{60} + R_{40}$

c) $R_{100} > R_{60} > R_{40}$ d) $\frac{1}{R_{100}} > \frac{1}{R_{60}} > \frac{1}{R_{40}}$

Solution

$$R_{100} = \frac{V^2}{100} \quad R_{60} = \frac{V^2}{60} \quad R_{40} = \frac{V^2}{40}$$

Let supply voltage be 'V'

$\Rightarrow R_{40} > R_{60} > R_{100}$

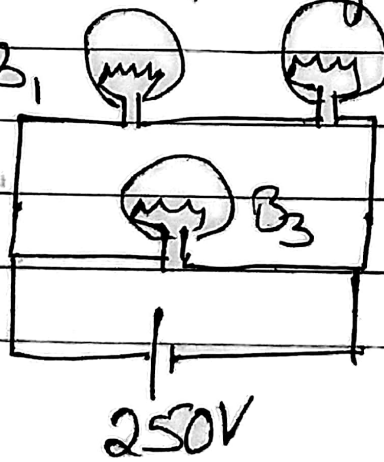
or $\frac{1}{R_{40}} < \frac{1}{R_{60}} < \frac{1}{R_{100}}$.

Q) IIT 2002

$B_1 \longrightarrow 100W$

B_2 & $B_3 \longrightarrow 60W$

power output of $B_1, B_2, B_3 = W_1, W_2, W_3$



a) $W_1 > W_2 = W_3$

b) $W_1 > W_2 > W_3$

c) $W_1 < W_2 = W_3$

d) $W_1 < W_2 < W_3$

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Solution

$$R_3 = \frac{V^2}{60}$$

Solution: $R_1 = \frac{V^2}{100}$ $\quad R_2 = R_3 = \frac{V^2}{60}$

Power in R_3 $\quad W_3 = \frac{(250)^2}{R_3}$

Power in combination of R_1 & R_2 $\quad W_1 + W_2 = \frac{(250)^2}{R_1 + R_2}$

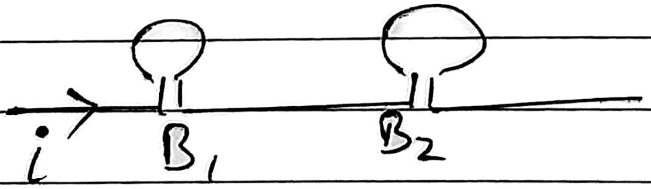
Clearly $(R_1 + R_2) > R_3$

$$\Rightarrow \frac{(250)^2}{R_1 + R_2} < \frac{(250)^2}{R_3}$$

$$\Rightarrow W_1 + W_2 < W_3$$

So W_3 is maximum

now compare W_1 & W_2



R_1 & R_2 in series current 'i' is sam

$W_1 = iR_1$ $\quad W_2 = iR_2$

clearly $R_2 > R_1$

$$\Rightarrow W_2 > W_1$$

So $W_3 > W_2 > W_1$

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